

Cloud Security Risk Assessment Report Governance, Risk, and Compliance (GRC)

Environment: Microsoft Azure

Virtual Machine: GRC-WIN-VM01

Operating System: Windows Server 2022 Datacenter

Assessment Type: GRC Non-Exploitative Security Assessment

1. Executive Summary

This report documents the results of a cloud security risk assessment conducted on a single Microsoft Azure-hosted virtual machine from a Governance, Risk, and Compliance (GRC) perspective. The purpose of the assessment was to identify critical assets, evaluate realistic threat scenarios, assess vulnerabilities, calculate risk scores, and recommend security controls aligned with recognized industry frameworks.

The assessment identified several medium-to-high risks, primarily related to publicly exposed Remote Desktop Protocol (RDP) access, the use of a privileged local administrator account without multifactor authentication, default operating-system hardening, and limited centralized monitoring and alerting. Windows Security logging and basic Azure monitoring were active; however, enhanced monitoring and security alerts were not configured.

No exploitation was performed during the assessment. The identified findings represent realistic cloud security risks that could be abused by attackers if they are not properly addressed.

2. Scope and Environment

Assessment Scope

- Single Azure virtual machine
- No additional cloud services or applications included
- Assessment performed from a GRC non-exploitative perspective

The assessment focused on a single virtual machine deployed in Microsoft Azure and configured to represent a small cloud-hosted business workload. The environment included a dedicated resource group, a Windows Server 2022 Datacenter operating system, public Remote Desktop access, a local administrative account, Azure networking resources, and built-in monitoring capabilities. Reviewing these components helped establish the scope of the assessment and provided the technical context needed to identify assets, evaluate threats and vulnerabilities, calculate risk, and recommend appropriate security controls. The table below summarizes the main configuration details of the Azure environment included in this assessment.

Environment Details

Item	Description
Cloud Provider	Microsoft Azure
Resource Group	GRC-Lab
Virtual Machine Name	GRC-WIN-VM01
Region	West US 2
Operating System	Windows Server 2022 Datacenter
VM Size	Standard B2als v2 — 2 vCPUs, 4 GiB memory

3. Asset Identification

The following assets were identified within the assessment scope. Assets include not only physical or virtual infrastructure, but also identities, network access points, operating-system components, cloud-management access, and security logs. Each asset was evaluated based on the value it provides and the potential impact if it were compromised or misused.

Asset Category	Asset	Why It Matters
Compute	Azure Virtual Machine	Hosts the operating system, data, and services
Identity	Local Administrator Account	Provides full administrative control
Network	Public IP Address and RDP Port	Provides an internet-accessible entry point
Cloud Control	Azure Subscription	Provides management and billing access
Operating System	Windows Server Services	May be vulnerable if not patched or hardened
Logging	Windows Security Event Logs	Records security activity used for detection and investigation

4. Threat and Vulnerability Summary

This section connects each identified asset to a realistic threat scenario and the vulnerability that could allow the threat to succeed. The findings are based on the configuration observed during the assessment, including public RDP access, administrative privileges, default operating-system security settings, and limited centralized monitoring.

Asset	Threat	Vulnerability	Why It Matters
Azure Virtual Machine	Brute-force attack	RDP exposed to the internet	Provides an easy entry point for automated attacks
Administrator Account	Credential compromise	No multifactor authentication	Could provide full administrative control
Operating System	Exploitation of vulnerabilities	Default security hardening	Common Windows vulnerabilities may be targeted
Security Logs	Undetected malicious activity	No centralized logging or alerts	Delays detection and incident response
Azure Subscription	Unauthorized configuration changes	Single privileged administrator	Misuse could affect high-impact cloud resources

5. Risk Assessment and Prioritization

Risks were evaluated using a standard likelihood-and-impact model to support consistent scoring and prioritization. Likelihood represents the probability that a threat could occur in the assessed environment, while impact represents the potential severity if the threat were successfully exploited.

Risk Scoring Model

- Likelihood: 1 (Low) to 5 (High)
- Impact: 1 (Low) to 5 (High)
- Risk Score = Likelihood × Impact

Asset	Threat	Likelihood	Impact	Risk Score	Justification
RDP	Brute-force attack	4	4	16	Public exposure
Administrator Account	Credential compromise	3	5	15	Full system control
Operating System	Unpatched vulnerabilities	3	4	12	Common Windows attack target
Azure Subscription	Privilege misuse	2	5	10	High-impact cloud resource
Security Logs	Undetected attack	3	3	9	Delayed detection and response

6. Security Controls and Compliance Mapping

The identified risks were mapped to preventive, detective, and corrective security controls aligned with recognized cybersecurity frameworks. This mapping demonstrates how each recommended control reduces a specific risk while supporting governance, auditability, and compliance requirements.

Risk	Security Control	Control Type	Framework
RDP attacks	Restrict inbound IP access	Preventive	CIS Control 4
Credential theft	Enforce MFA	Preventive	NIST CSF PR.AC
Operating-system exploits	Patch management	Corrective	CIS Control 7
Lack of centralized logging	Centralized logging	Detective	NIST CSF DE.CM
Privilege misuse	RBAC enforcement	Preventive	CIS Control 5

7. Recommended Mitigation Actions

The following actions are recommended to reduce the identified risks and strengthen the security of the Azure environment:

- Restrict inbound RDP access through the Network Security Group to trusted IP addresses only.
- Enable multifactor authentication for all administrative and privileged accounts.
- Apply operating-system patches and security updates regularly.
- Enable centralized logging and security alerts through Azure Monitor or Microsoft Defender for Cloud.
- Implement role-based access control to enforce least privilege for Azure resources.
- Periodically review administrative accounts, access permissions, and network security rules.

8. Overall Risk Posture

Based on the findings, the overall risk posture of the assessed Azure environment is **Medium to High**. The virtual machine is limited in scope, but publicly exposed RDP access creates a significant attack surface and increases the likelihood of automated brute-force attempts. The local administrator account provides full system control, which increases the potential impact of credential compromise.

Windows Security event logging and basic Azure monitoring were active during the assessment. However, centralized logging, enhanced monitoring, automated security alerts, multifactor authentication, and restricted RDP source access were not configured. These gaps could delay the detection of malicious activity and increase the impact of unauthorized access.

The environment's risk level can be reduced by restricting RDP access to trusted IP addresses, enabling multifactor authentication, applying regular patches, enforcing least privilege through RBAC, and implementing centralized monitoring and alerting.